Chia-Yuan Chiang

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Education

Yuan Ze University — Taoyuan, Taiwan
B.S. in Electrical Engineering — *Sep. 2024 – Sep. 2027 (Expected)*

Ming Chi University of Technology — New Taipei, Taiwan
B.S. in Electrical Engineering (Transferred) — *Sep. 2023 – Jun. 2024*

Nangang Senior High School — Taipei, Taiwan
Electronics Technology — *Sep. 2020 – Jun. 2023*

Experience

Undergraduate Researcher — May 2025 – Present
Prof. Shu-Yen Lin's Lab, Yuan Ze University — *Taoyuan, Taiwan*

- Led an early-warning model for ICU Ventilator-Associated Pneumonia (VAP); designed a stay-level 5-fold cross-validation scheme to eliminate patient-level data leakage present in the public benchmark.
- Applied Integrated Gradients feature attribution to reduce the model to 4 non-invasive features; the resulting LSTM reached 0.98–0.99 AUROC at a 72-hour prediction horizon.
- Manuscript submitted to IEEE GCCE 2026.

Engineering Advisor & Systems Integrator — 2023 – Present
FRC Team 7645 — *Taipei, Taiwan*

- Own robot system architecture: multi-sensor fusion, PID closed-loop motor control tuning, and software/hardware interface definitions.
- Introduced LLM-assisted tooling that cut iterative firmware debugging time by roughly one week; trained junior members to debug from root-cause analysis instead of trial-and-error.
- Rebuilt the team website, reducing repetitive PR/outreach inquiries.

Competitor — Jun. 2020 – Sep. 2023
Nangang Senior High School – NK-MTC 7645 — *Taipei, Taiwan*

- Generalist across LabVIEW/Arduino firmware, FRC electrical wiring, AutoCAD modeling, and 3D-printed mechanical parts; competed in 3 FRC regional events and represented the team at the National Skills Competition.

Projects

ICU VAP Early-Warning Model | *PyTorch, LSTM, SHAP* — 2025 – Present

- Single-center pilot (Far Eastern Memorial Hospital ICU, n=109); designed stay-level 5-fold stratified CV that exposed patient-level data leakage in the PREDICT 2025 benchmark (AUROC dropped from an inflated 0.99 to a real 0.58).
- Used Integrated Gradients to drop a collinear redundant feature ($\text{mv\_total} \equiv \text{Vt} \times \text{RR}$) and reduced the model to 4 non-invasive features; final LSTM reached 0.99 AUROC at a 24-hour horizon, 0.98–0.99 across 6–72h.

Open-Source QMK x STM32 Numpad | *C, ChibiOS HAL, EasyEDA* — 2026

- Designed a 17-key PCB in EasyEDA Pro, hand-soldered the board, and flashed STM32F072 via DFU; reverse-engineered ChibiOS HAL config (chconf/halconf/mcuconf) to resolve build conflicts and timing issues between the open-source PCB and QMK firmware.
- Reached stable USB HID enumeration with ¡5ms key latency, working RGB Matrix, and live VIA/VIAL remapping without recompiling.

Swerve Drive Chassis (FRC) | *LabVIEW, CNC, PID Control* — 2020 – 2023

- Designed and machined 3 generations of a swerve-drive chassis from scratch (kinematics, gear train, CNC/wire-EDM fabrication, LabVIEW PID control); won Innovation in Control Award at 2022 FRC Taiwan Hon Hai Regional.

Technical Skills

Languages & Firmware: C/C++ (STM32/Cortex-M), Python, QMK/ChibiOS HAL, LabVIEW, Arduino/ESP32, UART/SPI/I2C
Machine Learning: PyTorch, TensorFlow/Keras, LSTM/Time-Series, Feature Engineering, Signal Processing, SHAP/XAI, TinyML
EDA / Hardware: Altium Designer, KiCAD, OrCAD, EasyEDA Pro
Manufacturing: CNC Milling, Mastercam, 3D Printing, Laser Cutting, Welding, AutoCAD, Fusion 360, Autodesk Inventor

Awards & Certifications

2022 FRC Taiwan Hon Hai Regional – Finalist Alliance & Innovation in Control Award · 52nd National Skills Competition – Team Project Category Representative (2022) · 2020 FRC CIT 5G Regional – Excellence in Engineering (Industrial Design) Award